CIS 371 Web Application Programming

ExpressJS & Cloud Functions

HTTP server in (Java|Type)Script

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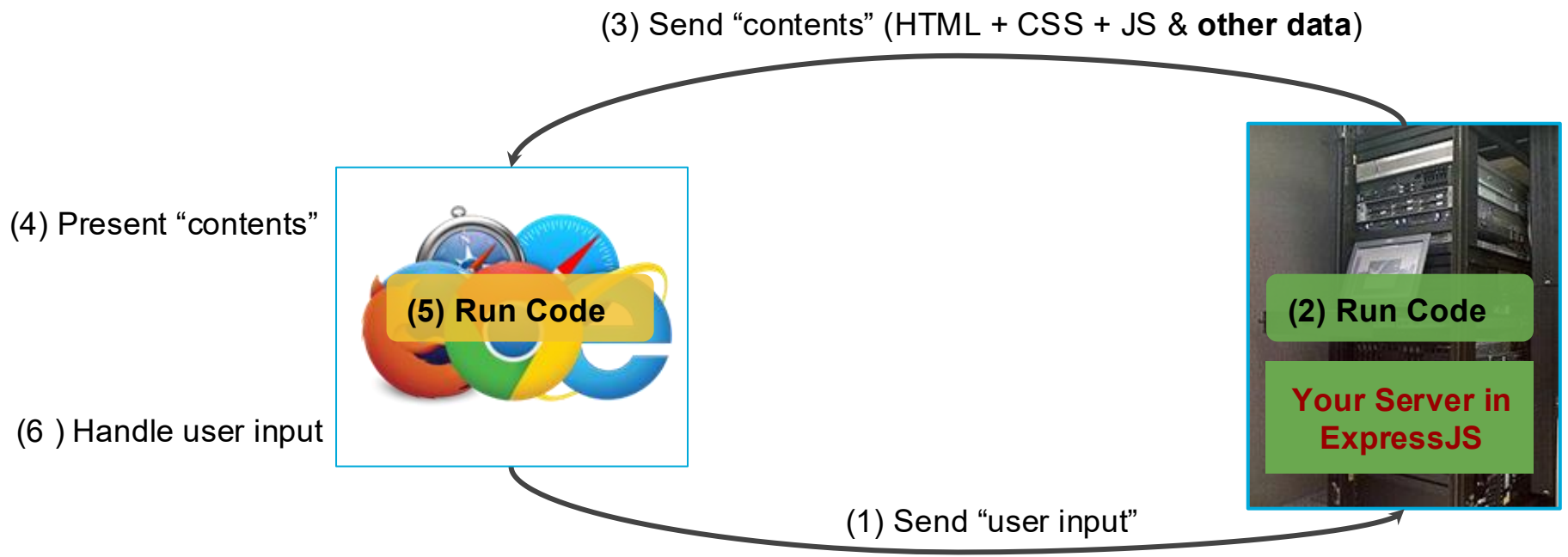
Based on the original version by Professor Hans Dulimarta and Professor Yong Zhuang.

Topics

- What is ExpressJS
- Why need an app server
- Features provided
  - Router for HTTP methods (GET, POST, PUT, DELETE, …)
  - Middleware
  - Query Parameter parsing
  - Payload parsing (via middleware)
- Prerequisite
  - HTTP Protocol: Request & Response

Web Client/Server Architecture



Why Use an App Server

- Enhanced Computing Power:
  - Client Side: Limited computing capabilities on user machines.
  - App Server: Typically hosted on robust external machines offering greater computing power.
- Security Considerations:
  - Code Visibility: Code sent to a client's browser can be inspected by the user, possibly revealing proprietary information.
  - Company Secrets: Using an app server can protect sensitive logic and data from direct client access.
- Be aware of extra network transport overhead
- Potential Use Cases:
  - Proxy Server: Bypass CORS restrictions when accessing certain web services.
  - Data Aggregation: Consolidate data from various sources like web clients and mobile clients.
  - High-End Computations: Ideal for tasks that require heavy computation, such as machine learning, computational simulations, and data mining.

Alternatives to ExpressJS

Setup

```
npm init -y
npm i express # version 4.x
```

```
// Additional libraries for TypeScript dev
npm i --save-dev typescript ts-node nodemon

// Add type declaration files
npm i --save-dev @types/express @types/node

// Creates tsconfig.json
tsc --init
```

tsconfig.json

```
{
  "compilerOptions":{
  "target": "es6",
  "module": "commonjs",
  "rootDir": "./",
  "outDir": "./build",
  "esModuleInterop": true,
  "strict": true
  }
 };
```

Hello World

```ts
import express, { Application, Request, Response } from "express";

const app: Application = express();
const PORT = process.env.PORT ?? 8000;

// Define GET endpoint(s)
app.get("/", (req: Request, res: Response) => {
  res.send("Hello World!");
});

app.listen(PORT, () => {
  console.log(`Server is listening to port ${PORT}`);
});
```

my-server.ts

```
// Launch the server
npx nodemon my-server.ts
```

Browser http://localhost:8000/

Defining More Routes/EndPoints

```ts
// import lines not shown
app.get("/", (req: Request, res: Response) => {
  res.send("Hello World!");
});
app.get("/about", (req: Request, res: Response) => {
  res.send("Just a simple Express server");
});
app.post("/xyz", (req: Request, res: Response) => {
  res.send("Just a simple Express server");
});
app.put("/order/cancel/", (req: Request, res: Response) => {
  res.send("_____");
});
app.delete("/account", (req: Request, res: Response) => {
  res.send("_____");
});
```

my-server.ts

Browser http://localhost:8000/

Browser http://localhost:8000/about

Can't invoke these from browser omnibox

Sending Responses

```
// METHOD: get, post, put, delete, and so on
app.METHOD("/path/goes/here", (req: Request, res: Response) => {
  // any file type, its content will be included in the response body
  res.download("file-name"); // Without MIME type
  res.type("image/jpg").download("mycat.jpg"); // OR with MIME type
});
```

Opt #1: file attachment

```
app.METHOD("/path/goes/here", (req: Request, res: Response) => {
  res.send("some text here"); // Opt-2: send text response
  res.type("text/plain").send("some text here"); // Opt-2a: with Content-Type
});
```

Opt #2: plain text

```
app.METHOD("/path/goes/here", (req: Request, res: Response) => {
  res.send("<P>Sample HTML response</p>"); // Opt-3: send HTML response
  res.type("text/html").send("<P>Sample HTML response</p>"); // Opt-3a: with Content-Type
});
```

Opt #3: HTML string

Sending Responses

```
app.METHOD("/path/goes/here", (req: Request, res: Response) => {
  res.send({ shippingCost: 5.16, sameDay: false }); // Opt-4: send JSON response
});
```

Opt #4: JSON

```
app.METHOD("/path/goes/here", (req: Request, res: Response) => {
  // Opt-5: send JSON from a JS object
  const my_resp_obj = { msg: "Hello World", hasEmoji: false };
  res.json(my_resp_obj);
});
```

Opt #5: JSON from object

```
app.METHOD("/path/goes/here", (req: Request, res: Response) => {
  res.status(401).end(); // Opt-6: set HTTP status & an empty response
});
```

Opt #6: empty response

Parsing Dynamic Path/Route Names

```
app.get("/user/:uname", (req: Request, res: Response) => {
  const who = req.params.uname;
  if (who == "___") res.send(`Option #1 here`);
  else res.send(`Option #2 here`);
});
```

Browser http://localhost:8000/user/**bob**

```
app.get("/search/:minPrice/:maxPrice", (req: Request, res: Response) => {
  const low = req.params.minPrice;
  const high = req.params.maxPrice;
  res.send(`Price range ${low}-${high}`);
});
```

Browser http://localhost:8000/search/**50/110**

Parsing Query Parameters (TypeScript types)

```
type Query2 = {
  min: number;
  max: number;
};
```

Browser http://localhost:8000/search?min=50&max=110

```
app.get("/search", (req: Request<any, any, any, Query2>, res: Response) => {
  res.send(`Price range ${req.query.min}-${req.query.max}`);
});
```

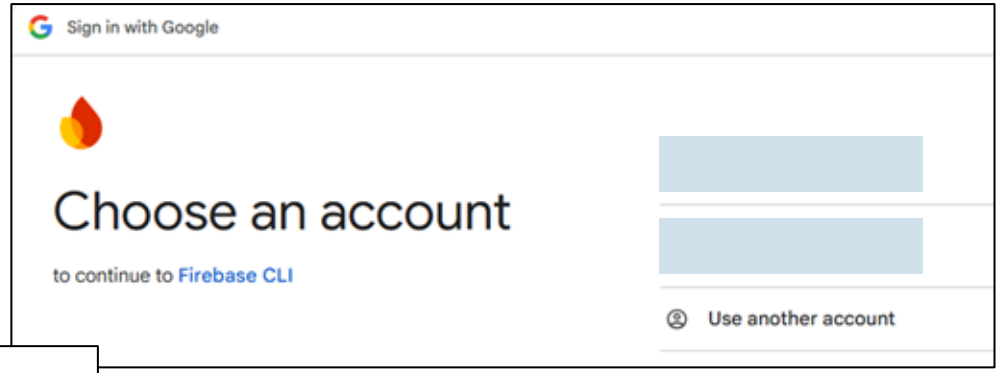
Recall: Firebase is a Collection of Many Products

- Authentication
- Cloud Firestore
- Cloud Storage
- **Cloud Functions**
- …

Project Setup & Initialization

```
node ➜~/projects $ npm install -g firebase-tools
```

```
node ➜~/projects $ firebase login
```



Woohoo!

Firebase CLI Login Successful

You are logged in to the Firebase Command-Line interface. You can immediately close this window and continue using the CLI.

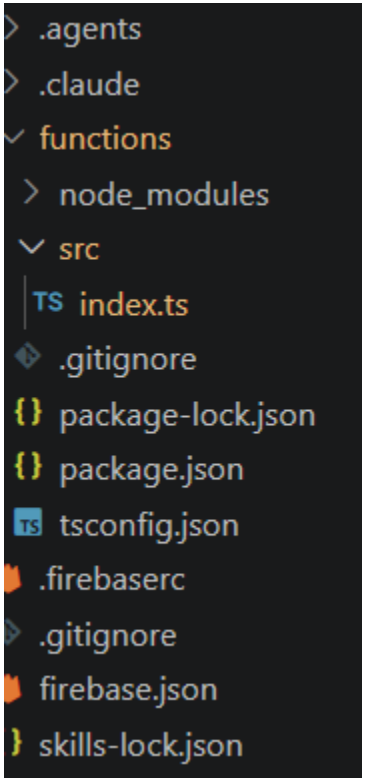
Setup & Initialization

```
node →~/projects $ mkdir fire-functions
node →~/projects $ cd fire-functions/
```

```
node →~/projects/fire-functions $ firebase init

     ######## #### ########  ######## ########     ###    ######  ########
     ##        ##  ##     ## ##       ##     ##  ##   ##  ##       ##
     ######    ##  ########  ######   ########  #########  ######  ######
     ##        ##  ##    ##  ##       ##     ## ##     ##       ## ##
     ##       #### ##     ## ######## ########  ##     ##  ######  ########

You're about to initialize a Firebase project in this directory:

  /home/node/projects/fire-functions

? Which Firebase features do you want to set up for this directory? Press Space to select features, then Enter to confirm your choices.
 ◯ Data Connect: Set up a Firebase Data Connect service
 ◯ Firestore: Configure security rules and indexes files for Firestore
 ◯ Genkit: Setup a new Genkit project with Firebase
❯◉ Functions: Configure a Cloud Functions directory and its files
 ◯ App Hosting: Set up deployments for full-stack web apps (supports server-side rendering)
 ◯ Hosting: Set up deployments for static web apps
 ◯ Storage: Configure a security rules file for Cloud Storage
```

```
✓ What language would you like to use to write Cloud Functions? TypeScript
✓ Do you want to use ESLint to catch probable bugs and enforce style? No
✓  Wrote functions/package.json
✓  Wrote functions/tsconfig.json
✓  Wrote functions/src/index.ts
✓  Wrote functions/.gitignore
✓ Do you want to install dependencies with npm now? Yes
```

Setup & Initialization



index.ts

```
 * See a full list of supported triggers at https://firebase.google.com/docs
 */

import {setGlobalOptions} from "firebase-functions";
import {onRequest} from "firebase-functions/https";
import * as logger from "firebase-functions/logger";

// Start writing functions
// https://firebase.google.com/docs/functions/typescript

// For cost control, you can set the maximum number of containers that can be
// running at the same time. This helps mitigate the impact of unexpected
// traffic spikes by instead downgrading performance. This limit is a
// per-function limit. You can override the limit for each function using the
// `maxInstances` option in the function's options, e.g.
// `onRequest({ maxInstances: 5 }, (req, res) => { ... })`.
// NOTE: setGlobalOptions does not apply to functions using the v1 API. V1
// functions should each use functions.runWith({ maxInstances: 10 }) instead.
// In the v1 API, each function can only serve one request per container, so
// this will be the maximum concurrent request count.
setGlobalOptions({ maxInstances: 10 });

// export const helloWorld = onRequest((request, response) => {
//   logger.info("Hello logs!", {structuredData: true});
//   response.send("Hello from Firebase!");
// });
```

Setting up End Points

```
export const api = onRequest(async (req, res) => {
  switch (req.method) {
    case "GET":
      res.send("GET request received!");
      break;
    case "POST":
      const body = req.body;
      res.send(body);
      break;
    case "DELETE":
      res.send("DELETE request received!");
      break;
    default:
      res.send("It was a default request!");
  }
});
```

index.ts

Install Firebase Emulators

```
node →~/projects/fire-functions $ firebase init emulators
```

```
i  Using project beverageshop-876a8 (BeverageShop) .

=== Emulators Setup
? Which Firebase emulators do you want to set up? Press Space to select emulators, then Enter to confirm your choices.
 ◯ App Hosting Emulator
 ◉ Authentication Emulator
 ◉ Functions Emulator
❯◉ Firestore Emulator
 ◯ Database Emulator
 ◯ Hosting Emulator
 ◉ Pub/Sub Emulator
```

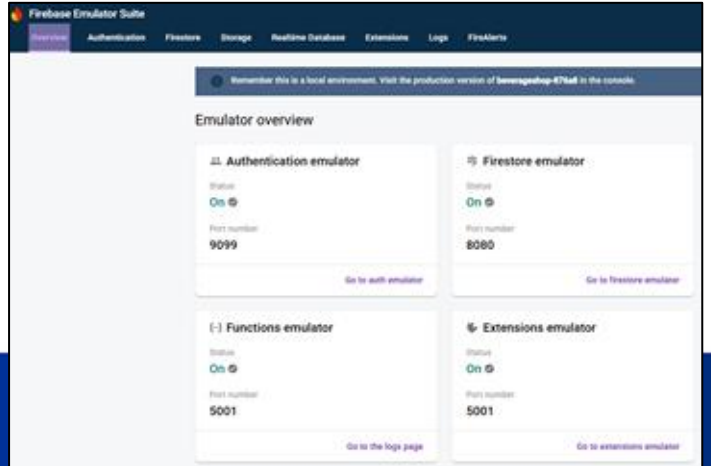
```
=== Emulators Setup
✓ Which Firebase emulators do you want to set up? Press Space to select emulators, then Enter to confirm your choices.
✓ Which port do you want to use for the auth emulator? 9099
✓ Which port do you want to use for the functions emulator? 5001
✓ Which port do you want to use for the firestore emulator? 8080
✓ Which port do you want to use for the pubsub emulator? 8085
✓ Would you like to enable the Emulator UI? Yes
✓ Which port do you want to use for the Emulator UI (leave empty to use any available port)?
✓ Would you like to download the emulators now? Yes
i  firestore: downloading cloud-firestore-emulator-v1.20.4.jar...
Progress: ========================================================================
i  pubsub: downloading pubsub-emulator-0.8.29.zip...
Progress: ========================================================================
i  ui: downloading ui-v1.15.0.zip...

=== Agent Skills Setup
If you are using an AI coding agent, Firebase Agent Skills make it an expert at Firebase.
Progress: ========================================================================
```

Firebase Emulators require JAVA

```
node ➜~/projects/fire-functions $ sudo apt-get update
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9,671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5,412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [123 kB]
Fetched 10.0 MB in 9s (1,097 kB/s)
Reading package lists... Done
node ➜~/projects/fire-functions $ sudo apt-get install -y openjdk-21-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

```
node ➜~/projects/fire-functions $ firebase emulators:start

✓ All emulators ready! It is now safe to connect your app.
i View Emulator UI at http://127.0.0.1:4000/
```

Emulator	Host:Port	View in Emulator UI
Authentication	127.0.0.1:9099	http://127.0.0.1:4000/auth
Functions	127.0.0.1:5001	http://127.0.0.1:4000/functions
Firestore	127.0.0.1:8080	http://127.0.0.1:4000/firestore
Pub/Sub	127.0.0.1:8085	n/a
Extensions	127.0.0.1:5001	http://127.0.0.1:4000/extensions



Firebase Emulators require JAVA

```
export const helloWorld = onRequest((request, response) => {
  logger.info("Hello logs!", { structuredData: true });
  response.send("Hello from Firebase!");
});
```

index.ts

```
● node →~/projects/fire-functions/functions $ npm run build

  > build
  > tsc
```

http://localhost:5001/<your-project-id>/us-central1/helloWorld

ID HPC Faculty Teaching YouTube Maps Gr

Hello from Firebase!